

MATH 415-01 Advanced Linear Algebra
Homework #1
Assigned: January 12, 2026
Due: January 21, 2026

You may use technology to row reduce matrices.

1. Do the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -2 \\ 4 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

span $\mathbb{R}^2$? There are multiple ways to answer this question, but use the definition. Show all of your work.

2. Are the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 5 \\ -3 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 1 \\ -2 \end{bmatrix}$$

linearly independent? There are multiple ways to answer this question, but use the definition. Show all of your work.

3. Are the polynomials

$$p_1(x) = 4 + 3x, \quad p_2(x) = 1 + x + 3x^3, \quad p_3(x) = 7 + 8x + x^2, \quad p_4(x) = -2x + 2x^3$$

a basis for P_3 ? There are multiple ways to answer this question, but use the definitions. Show all of your work. Do not simply throw numbers into a matrix. Work out the steps.

4. Determine whether or not the following sets are subspaces of $\mathbb{R}^3$.

(a)

$$S_1 = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} : x_1 + x_3 = 0 \right\}$$

(b)

$$S_2 = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} : x_1 = 0 \text{ or } x_3 = 0 \right\}$$

5. Find the eigenvalues and eigenvectors for

$$D = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}.$$

Show all of your work.

6. Find a basis for each of the eigenspaces for

$$E = \begin{bmatrix} 5 & 1 & -2 \\ -1 & 3 & 2 \\ 1 & 1 & 6 \end{bmatrix}.$$

Show all of your work. Also state the algebraic and geometric multiplicities of each eigenvalue.